

Biologist and aquatic scientist interested in phytoplankton and other microbial eukaryotes, as well as the intersection between aquatic ecosystems and human health.

RESEARCH AND WORK EXPERIENCE

Postdoctoral Investigator, Alexander Lab, Woods Hole Oceanographic Institution, MA **2026 – present**

- Investigating population connectivity and dispersal of dinoflagellates by leveraging meta-transcriptomic and single-cell transcriptomic data.

Doctoral Research Assistant, Brosnahan Lab, Woods Hole Oceanographic Institution, MA **2021 – 2026**

Dissertation: “Mechanisms Driving Development and Persistence of Inshore *Dinophysis* Blooms”

- **Life cycle dynamics:** discovered sexual reproduction in *Dinophysis* is rapid, associated with vegetative growth, and drives most of bloom proliferation.
- **Metabolic-behavioral linkages:** determined that changes in metabolic mode of the mixotrophic *Dinophysis* influenced vertical swimming and thin layer dynamics.
- **Physical-ecological interactions:** demonstrated how coastal hydrography and physico-chemical niche stability in kettle ponds during blooms promotes persistence of *Dinophysis* populations.
- **Population connectivity:** developed a novel metatranscriptomic SNP genotyping method to identify genetic connectivity between dinoflagellate bloom populations.
- Organized coastal field sampling campaigns, designed and conducted lab culture experiments, developed molecular methods, and deployed diverse computational tools (CNN classifiers, bioinformatic pipelines).

In addition to dissertation work:

- Led or assisted with field sampling efforts on the Cape Cod National Seashore, MA to deploy or recover instruments, and sample for nutrients, plankton community, and toxins (about 8-10 times per year).
- Developed a convolutional neural net (CNN) model for phytoplankton image identification and evaluated 3-4 other CNN models for five HAB species across seven locations.
- Designed and coded new applications for a sensor-coordinating robot operating system (ROS).
- Developed high-throughput single-cell RNA sequencing approach for microbial eukaryotes.
- Mentored 4-5 students (undergraduate and graduate), including being primary mentor for a summer undergraduate student in 2025.

Analyst, Cadmus Group, LLC, Waltham, MA **2019 – 2021**

For U.S. EPA Office of Water clients:

- developed national physical and social watershed vulnerability indicators for states and other stakeholders
- created pipeline for summarizing health effects and occurrence of 250 drinking water contaminants for the regulatory Contaminant Candidate List
- produced maps and graphs on sea-level rise and extreme weather events for EPA tools
- moderated trainings for water utilities on EPA emergency resilience rules

Research Assistant, Vallen Lab, Swarthmore College, PA **2017 – 2019**

- Developed project identifying novel immune protein homologues in the sea anemone *Aiptasia pallida* using transcriptomic and molecular methods to understand the cellular processes behind coral bleaching.

EDUCATION

Ph.D.	MIT-WHOI Joint Program in Oceanography/Applied Ocean Sciences and Engineering, Biological Oceanography	July 2026
B.A.	Swarthmore College, Biology and Political Science High Honors, 3.89 GPA	May 2019

PUBLICATIONS

Sung-Clarke, S., N. Ayache, W. Zhang, M. Tong, J. L. Smith, and M. Brosnahan. 2026. Rapid sexual reproduction intensifies blooms of a mixotrophic dinoflagellate. *Limnology and Oceanography* 71:e70419

Sung-Clarke, S., N. Ayache, W. Zhang, D. Ralston, E. Lechner, C. Roesler, S. Drapeau, J.L. Smith, M. Tong, A. Wang, & M. Brosnahan. Cessation of diel vertical migration by an inshore dinoflagellate bloom during prey deprivation. [in review]

SERVICE AND LEADERSHIP

Teaching Fellow, Biological Oceanography, Massachusetts Institute of Technology, MA **Spring 2025**

- Instructed weekly 1-hour recitation sessions reviewing course content for the class.
- Developed and presented a lecture on marine protists.
- Graded project papers, quick writes, and problem sets for seven students.

MIT-WHOI Student Government Biology Representative, Woods Hole, MA **2023 – 2024**

- Organized student socials and bimonthly Biology Department coffee hours.
- Hosted ocean biologist Dr. Kelly Benoit-Bird for two days as the 2024 Steinbach Scholar, organized four speaking events and set up ~20 meetings around her visit.
- Organized the Joint Program Annual Meeting with the Academic Programs Office.
- Created and led practice qualifying exam review sessions for pre-candidacy students in 2024 and 2025.

Broader Impacts Group Leader, Woods Hole, MA **2022 – 2025**

- Liaised between outreach coordinators and students to match students to volunteer opportunities.
- Volunteered with several field trips, science nights, and science fairs.
- Trained WHOI outreach team on use of folding paper microscopes for school activities.

Gordon Research Seminar Discussion Leader, Easton, MA **June 2025**

- At the Mycotoxins and Phycotoxins Gordon Research Seminar, moderated discussion and questions for the session entitled “Environmental Factors Influencing Toxin Production”.

PRESENTATIONS

Sung-Clarke, S., N. Ayache, W. Zhang, D. Ralston, E. Lechner, C. Roesler, S. Drapeau, J.L. Smith, M. Tong, A. Wang, & M. Brosnahan (May 2026). “Low pH associated with thin layers of the mixotrophic dinoflagellate *Dinophysis*” [Poster Presentation]. NOAA Ocean Acidification Program Meeting, College Park, MD.

Sung-Clarke, S., N. Ayache, W. Zhang, D. Ralston, E. Lechner, C. Roesler, S. Drapeau, J.L. Smith, M. Tong, A. Wang, & M. Brosnahan (April 2026). “Coastal features on Cape Cod trap and intensify *Dinophysis* blooms” [Oral Presentation]. Gulf of Maine Harmful Algal Bloom Symposium, Portsmouth, NH.

Sung-Clarke, S., N. Ayache, W. Zhang, D. Ralston, E. Lechner, C. Roesler, S. Drapeau, J.L. Smith, M. Tong, A. Wang, & M. Brosnahan (February 2026). “Vertical swimming drives coastal trapping of the mixotrophic dinoflagellate *Dinophysis*” [Poster Presentation]. Ocean Sciences Meeting, Glasgow, Scotland.

Sung-Clarke, S., N. Ayache, W. Zhang, M. Tong, J.L. Smith, & M. Brosnahan (October 2025). “Rapid sexual reproduction intensifies and prolongs blooms of *Dinophysis acuminata*” [Oral Presentation]. International Conference on Harmful Algae, Punta Arenas, Chile. [One of 10% of presentations nominated for publication in special issue of *Harmful Algae*].

Sung-Clarke, S., N. Ayache, W. Zhang, M. Tong, J.L. Smith, & M. Brosnahan (April 2025). “Time partitioning of grazing, division, and mating promotes rapid sexual reproduction by a mixotrophic dinoflagellate” [Seminar Presentation]. Woods Hole Oceanographic Institution, Woods Hole, MA.

Sung-Clarke, S., N. Ayache, W. Zhang, M. Tong, J.L. Smith, & M. Brosnahan (March 2025). “Time partitioning of grazing, division, and mating by a mixotrophic dinoflagellate” [Oral Presentation]. Gulf of Maine Harmful Algal Bloom Symposium, Portsmouth, NH.

Sung-Clarke, S., N. Ayache, W. Zhang, J.L. Smith, & M. Brosnahan (October 2024). “Linked phasing of division and mating in *Dinophysis acuminata* culture and field populations” [Oral Presentation]. U.S. Symposium on Harmful Algae, Portland, ME.

Sung-Clarke, S., M. Pathare, V. Lucke, J.L. Smith, & M. Brosnahan (November 2023). “It’s a Trap! Biophysical trapping and phased mating during a *Dinophysis acuminata* bloom in Nauset Marsh.” [Oral Presentation]. International Conference on Harmful Algae, Hiroshima, Japan. [Winner of Maureen Keller Best Student Presentation Award].

Sung-Clarke, S., M. Pathare, V. Lucke, J.L. Smith, & M. Brosnahan (June 2023). “Biophysical trapping as a driver of *D. acuminata* bloom development and persistence in Nauset Marsh” [Oral Presentation]. Phycological Society of America Annual Meeting, Providence, RI.

Sung-Clarke, S., M. Pathare, D. Beaudoin, H. Alexander, & M. Brosnahan. (October 2022). “Investigating *Dinophysis* response to prey scarcity in Nauset Marsh” [Presentation and Lightning Talk]. U.S. Symposium on Harmful Algae, Albany, NY.

Sung-Clarke, S. (May 2019). “Identifying TIR domains and protein-protein interactions in the sea anemone *Aiptasia pallida*” [Honors Thesis Defense]. Defended to Dr. Angela Poole, Swarthmore College, PA.

Sung-Clarke, S. (May 2019). “Taiwan’s Wetland Conservation Act: A tactic for maintaining international space” [Oral Presentation]. Swarthmore College Political Science Thesis Symposium, Swarthmore College, PA.

AWARDS AND FELLOWSHIPS

Nominee, Invite-Only Special Issue, *Harmful Algae*, 2025. One of 35 selected from over 300 presentations at the International Conference on Harmful Algae 2025 for the special issue: “Emerging Insights and Perspectives on Harmful Algal Blooms 2026”.

Grassle Fellowship, Woods Hole Oceanographic Institution, 2024. \$7500 grant to support biodiversity and biophysical research activities for dissertation work.

Maureen Keller Best Student Presentation Award, International Conference on Harmful Algae, 2023.

CERTIFICATES

Research Mentorship Certificate, MIT, 2024. 3-week certificate program learning best practices for mentoring research students.

Grant Writing Certificate, MIT, 2024. 2-day certificate program in writing grants to federal agencies.

Subject Design Workshop, MIT, 2024. 3-week workshop to design course curricula.

Lesson Planning Workshop, MIT, 2026. 2-week workshop course to learn to plan active-learning lessons.

TECHNICAL SKILLS

Lab and Instrument Skills: Light microscopy, flow cytometry, Imaging FlowCytobot, CTDs, RNA/DNA library preparation

Programming: HPC/SLURM Computing; Bash; Python; Snakemake; R; Jupyter; GitHub; ArcGIS; Excel VBA

Applications: Microsoft Office, PowerBI; Adobe Illustrator, Adobe Photoshop, Adobe Captivate